DO Prediction Model for the Han River

Hourly dissolved oxygen forecasting using 17,965 observations and PatchTST (Transformer), with strong out-of-sample performance on unseen future data.

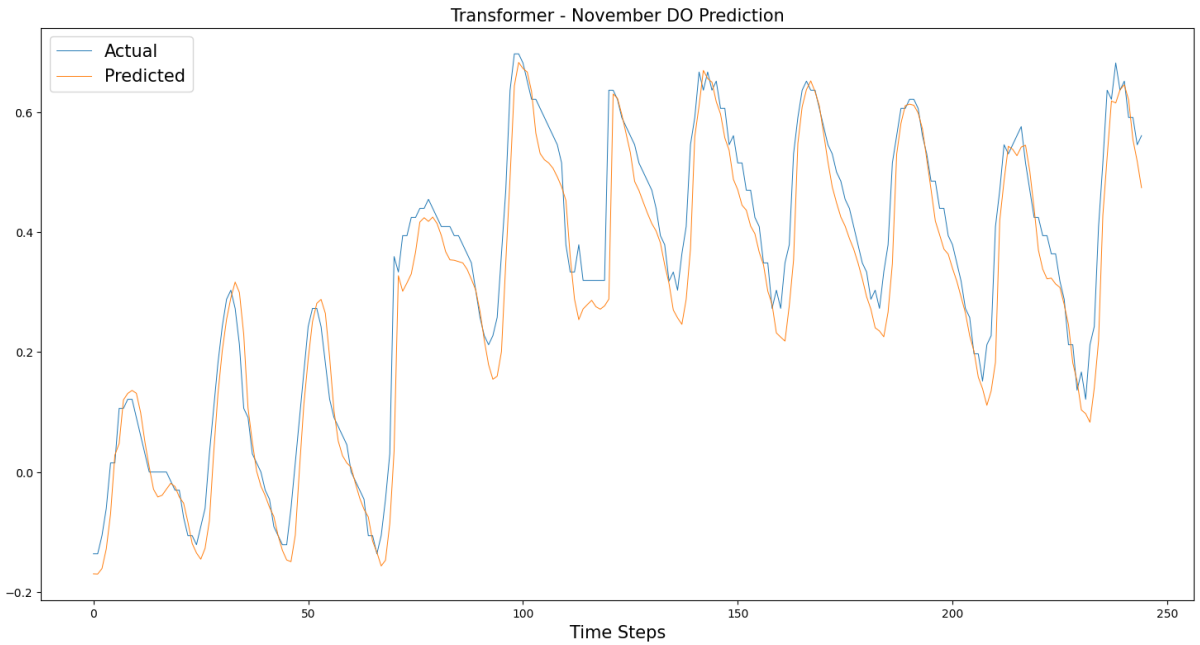

Figure 1. Out-of-sample forecasting on unseen November 2024 data

Project Snapshot

- **Target:** Dissolved Oxygen forecasting
- **Site:** Han River
- **Data:** 17,965 hourly observations (2022–2024)
- **Inputs:** Temperature, pH, TN, TP, TOC
- **Preprocessing:** Ensemble imputation
- **Best model:** PatchTST
- **Best performance:** $R^2 = 0.970$, RMSE = 0.015

Table 1. Comparative Analysis of Performance Metrics for Deep Learning Architectures on Unseen Data

Model	MSE	RMSE	MAE	R^2
LSTM	0.00029	0.017	0.012	0.961
GRU	0.00084	0.029	0.015	0.915
PatchTST	0.00021	0.015	0.010	0.970

This workflow evaluated multiple missing-data imputation strategies and identified ensemble imputation as the most effective preprocessing approach. Among the tested deep learning models, PatchTST achieved the best generalization on held-out future data, capturing both short-term periodicity and longer-term temporal structure.